

REGULAR ARTICLE

Demographic genetic analyses of the American beech (*Fagus grandifolia* Ehrh.) I. Genetic substructurings of northern populations with root suckers in Quebec and Pennsylvania

KEIKO KITAMURA,* HIDEKI TAKASU,† KAZUHIKO HAYASHI,‡ MASASHI OHARA,
§ TOMOSHI OHKAWA,¶ FREDERICH H. UTECH** and SHOICHI KAWANO¶

*Forestry and Forest Products Research Institute, Department of Agriculture, Forestry and Fisheries, Tsukuba, Ibaraki 305–8687, Japan, †Department of Education, Wakayama University, Wakayama 640–8441, Japan, ‡Biological Laboratory, Osaka Gakuin University, Suita, Osaka 564–8511, Japan, §Department of Biology, Graduate School of Arts and Sciences, University of Tokyo, Komaba, Meguro-ku, Tokyo 153–8902, Japan, ¶Department of Botany, Graduate School of Science, Kyoto University, Kyoto 606–8502, Japan (Email: skawano@ip.media.kyoto-u.ac.jp), **Section of Botany, Carnegie Museum of Natural History, Pittsburgh, PA 15213, USA

Abstract

Allozyme analyses were conducted to investigate the genetic substructurings of the American beech populations with root sucker formation. A total of 1364 shoots from four study plots in three local populations in its northern distributional range were thoroughly mapped, their sizes (diameter at breast height or diameter at ground height) measured and genotyped for 10 enzyme systems, and then multilocus genotypes were identified for each individual shoot. All four study plots turned out to be multiclinal populations in which the average number of shoots per multilocus genotype ranged from 1.17 to 3.22. Clonal diversities for the overall population were as high as 0.91–0.99 for Simpson's index (D) and 0.83–0.98 for Fager's index of evenness (E). Wright's F -statistics showed positive significant deviations from the Hardy–Weinberg Equilibrium in intermediate size-classes, a phenomenon which is attributed to overlapping generations with identical genotypes within these size-classes. Radial patterns of root sucker formation were intensive in three study plots, and size advantages in the production of root suckers were evident. Spatial autocorrelation analyses (standard normal deviates and coancestry) revealed that the effective range of root suckers in the American beech was 10–15 m. Seed production of the northern population of the American beech generates multiclinality, and the spatial genetic substructurings of local populations are subjected to extensive root sucker formation.

Keywords: allozyme, *Fagus grandifolia*, F -statistics, root sucker, size-class discrimination, spatial autocorrelation.

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The evolutionary and ecological features as well as the genetic diversity of clonal plants have been addressed in many plant species, mostly in relation to the advantages and disadvantages of sexual versus asexual reproduction (see reviews: Ellstrand & Roose 1987; Widén *et al.* 1994;

McLellan *et al.* 1997). The clonality of long-lived woody plants reflects various types of growth habits and is related to diverse life-history strategies that are different from those of herbaceous plants owing to characteristics such as individual size, longevity and woodiness (Kawano 1975, 1985; Jenik 1994; Peterson & Jones 1997). Various aspects of clonality of woody plant species have been investigated (Peterson & Jones 1997); however,

Correspondence: Shoichi Kawano

demographic genetic structures of populations, which comprise vegetatively established ramets and sexually produced new genets, have not yet been fully investigated in woody species.

The American beech is a late-successional, shade-tolerant dominant tree and one of the major climax components of the hardwood and mixed forests of eastern and central North America (Fowells 1965). It reproduces by both seeds and root suckers (Ward 1961; Jones & Raynal 1988), and shows a considerable geographic variation in the proportion of seedlings and root suckers, apparently with an increase in the extent of root suckering in the northernmost and high-elevation localities of its geographic range (Held 1983; Jones & Raynal 1986).

The post-Pleistocene distributional history of the American beech was revealed by pollen analyses (Davis 1983; Bennet 1985; Delcourt & Delcourt 1987), which suggested that the American beech populations with root sucker formation advanced to the Appalachian highlands about 14 000 B. P. and also migrated into the northeastern glaciated territories of Quebec and Nova Scotia. Our previous study on the geographic genetic variation of the American beech populations clearly shows that the root sucker populations are adapted to northern and upland environments, and that it had retreated into the glaciated territories after the Pleistocene glaciation period (K. Kitamura & S. Kawano, unpublished observation, 1998).

By extensive root suckering, the American beech forms a number of aboveground shoots, which belong to the identical individual (genet). The root sucker forms more or less radial patterns of shoots around its mother trunk; the effective radius of the root sucker shoot has been reported to be approximately 8 m by field survey (Jones & Raynal 1986). It is difficult, however, to identify the clonality of shoots in the field populations when shoots originating from the root suckers are intermingled. Houston and Houston (1994) applied the isozyme analysis to identify clonality of the American beech populations with root sucker formation, and demonstrated the usefulness of isozymes as genetic markers.

We investigated three northern populations of the American beech showing extensive root sucker formation. Two populations in Quebec, Canada, are near the northernmost limit of its geographic range. Another population in Pennsylvania is located in the Great Lakes region, which is in the center of the present-day geographic range of the American beech (Bennet 1985; Delcourt & Delcourt 1987). Since most studies of clonal plants have not examined genet distributions on a scale small enough to enable assessment of the potential interaction among different genotypes (Widen *et al.* 1994), we attempted to introduce a thorough mapping, measuring, and sampling of all shoots within study plots of the American beech populations. Then we genotyped all of

these sampled individuals to reveal the genet and ramet substructurings found in local populations in fine-scale. We have also attempted to estimate levels of genetic variation based upon allozyme loci and to identify genets by multilocus genotypes determined by a combination of different loci (McLellan *et al.* 1997).

The objectives of this study are: (i) to determine levels of genetic diversity; (ii) to examine the contributions of sexually reproduced seedlings to the genetic variation; (iii) to examine the spatial structures of intermingled genets; and (iv) to clarify the spatio-temporal patterns of differentiation for local populations of the American beech with extensive root sucker formation in the vicinity of the northernmost range. A discussion was also attempted concerning the origin of a unique growth form of root sucker formation and demographic genetic substructurings in relation to the environmental constraints of the habitats of this important dominant element of the eastern North American hardwood forests.

Materials and methods

Study sites

We established four study plots in three localities of American beech: one transect (10×50 m) in Powdermill, PA, USA (PWD); one transect (5×45 m) in Mont St. Hilaire, Quebec, Canada (MSH); and two transects (5×120 m and 5×20 m) in Duchesnay, Quebec, Canada (Q1 and Q2) (Table 1). Root suckers of the American beech are dominant in these four study plots.

Powdermill is located in the Powdermill Field Station of Carnegie Museum, PA. The American beech stands are mainly established in the fluvial plain along the creek. The study plot (10×50 m) was established on the mesic to moist flat terrace.

The McGill University Field Station at Mont St. Hilaire is located in the St Lawrence lowlands of Southern Quebec, northeast of Montreal, Quebec, Canada (Maycock 1961; Burt & Bell 1991). The study plot (MSH; 5×45 m) was established on a flat fluvial plain of the Lac Hertel, adjacent to the glaciated bluff dominated by several maple and oak species (Maycock 1961). The co-occurring dominant woody species with the American beech are *Acer saccharum* (sugar maple) and *Acer pensylvanicum*.

Two transects (Q1 and Q2) were established in the south-facing slope above Lac aux Deux-Frutes, Station Forestiere de Duchesnay, northwest of Quebec City, Quebec, Canada. This forest is one of the northernmost populations of the American beech. A large transect (Q1: 5×120 m) was established on the steep slope (approximately 30°) and a small transect (Q2: 5×20 m) on the flat ground at the base of the same slope covered by rocky

Table 1 Genetic variations in four study plots in three local populations of the American beech with root sucker formation

Study plot	Plot size (m)	No. shoots sampled	Shoot density (per m ²)	No. MLG	No. PL	No. MLG (per m ²)	No. shoots /MLG	D*	E [†]
PWD	10×50	190	0.38	59	8	0.12	3.22	0.91	0.83
MSH	5×45	271	1.20	136	8	0.60	1.98	0.97	0.89
Q1	5×120	840	1.40	354	11	0.59	2.37	0.99	0.98
Q2	5×20	63	0.63	54	9	0.54	1.17	0.99	0.85

MLG, multilocus genotype; PL, polymorphic loci.

*Simpson's index of clonal diversity; [†]Fager's index of evenness.

soils. The American beech is mixed with the following woody species: *Taxus canadensis*, *A. saccharum*, *A. pensylvanicum* and *Betula alleghaniensis*. The Q2 population is substantially dominated by juveniles of *A. saccharum*, with beech juveniles extremely suppressed and scarcely distributed.

We mapped the locations and measured the sizes (diameter at breast height (d.b.h.) or diameter at ground height (d.g.h.) and height) of all shoots within the study plots and of mature American beech trees in the surroundings of the transect. The number of mapped shoots ranged from 63 in Q2 to 840 in Q1 (Table 1).

Allozyme electrophoresis

We collected leaf samples for allozyme electrophoresis from all mapped shoots and surrounding trees in May 1995 at MSH and in July 1996 at PWD, Q1, and Q2. Samples were put into an ice bath during the transportation to the laboratory and stored at -80°C until the enzymes were extracted.

Enzyme extraction (Shiraishi 1988) and polyacrylamide vertical slab gel electrophoresis (Davis 1964; Orstein 1964) were carried out using the same procedures described in K. Kitamura and S. Kawano (unpublished observation, 1998). Ten enzyme systems were used for detection of 13 polymorphic allozyme loci: *6Pgdh1*, *6Pgdh2*, *6Pgdh3*, *Acp1*, *Acp2*, *Adh1*, *Amy2*, *Fum*, *Got1*, *Lap*, *Pgi* and *Pgm* (K. Kitamura & S. Kawano, unpublished observation, 1998).

Data analysis

Size-class discrimination. All mapped shoots were divided into five size-classes according to the diameter at ground height (d.g.h.): class 1=d.g.h.<5 mm; class 2=d.g.h. between 5 and 10 mm; class 3=d.g.h. between 10 and 25 mm; class 4=d.g.h. between 25 and 300 mm; and class 5=d.g.h. over 300 mm. At Q2, we sampled a shoot 30 mm in diameter, which obviously derived from a root sucker, and counted 33 annual growth rings. Although it was a suppressed shoot, this result gives a tentative criterion for an age estimation of the size-class 3–4.

The genotypic, allelic, and zygotic differentiations among five size-classes were calculated using the Chi-squared test of independence at a critical probability of Bonferroni approach, $\alpha' = \alpha/k$, where k is the number of intended tests (Sokal & Rohlf 1995).

Clonal diversity. We calculated two measures of clonal diversity: Simpson's index (D) and Fager's genotypic evenness (E) for each size-class. For calculations, we identified a multilocus genotype as a clone.

Simpson's index (D : Pielou 1966), corrected for finite sample size, is most sensitive to changes or differences in the most common genotype:

$$D_{\text{observed}} = 1 - \frac{\sum_{i=1}^G n_i(n_i - 1)}{N_r(N_r - 1)}$$

where n_i is the number of shoots of multilocus genotype i ; N_r is the total number of shoots; and G is the observed number of multilocus genotypes.

The standardized measures of clonal diversity were obtained by Fager's genotypic evenness (E : Fager 1972). Evenness provides a correct measure of Simpson's index for the number of multilocus genotypes within a population. Evenness ranges from 0 to 1, where 1 is the maximum equitability. When D_{maximum} varies among populations, evenness provides a standardized measure that allows interpopulation comparison (Montalvo *et al.* 1997):

$$E = \frac{D_{\text{observed}} - D_{\text{minimum}}}{D_{\text{maximum}} - D_{\text{minimum}}}$$

$$D_{\text{minimum}} = \frac{(G-1)(2N_r - G)}{N_r(N_r - 1)}$$

$$D_{\text{maximum}} = \frac{N_r(G-1)}{G(N_r - 1)}$$

where D is Simpson's index.

F-statistics and Hamilton's relatedness. Wright's F -statistics and Hamilton's relatedness (R) were calculated between five size-classes within each study plot.

Wright's F -statistics (Wright 1943, 1951, 1965) were estimated based on Weir and Cockerham (1984). F_{ST} was cal-

culated among five size-classes for each transect, and F_{IS} was calculated for each size-class.

The average relatedness of shoots was measured by Hamilton's (1971) relatedness using Queller and Goodnight's (1988) estimator (R):

$$R = \frac{2F_{ST}}{1 + F_{IT}}$$

F -statistics and relatedness were calculated using the computer program FSTAT developed by Goudet (1995), in which levels of significance are obtained by jackknifing procedure.

Spatial autocorrelation. Two measures of spatial autocorrelation were employed. One is the standard normal deviate (SND) and the other is coancestry (r_{ij}). Distance classes for calculations were 5-m intervals for all populations. For the Q1 population, calculations were made for upper slope ($x < 88$ m) and lower slope ($88 \text{ m} < x$) separately based on shoot positions.

Standard normal deviate calculates nominal correlations of genotypes, which is the deviation of the observed number of genotype joins from the expected number of joins (Sokal & Oden 1978a,b). We simply calculated SND for identical genotype pairs (like-joins), from which we estimate the ranges of root suckers. Genotypes for which the number of shoots was fewer than 10 were excluded from calculation, except for the Q2 population, in which we excluded genotypes with fewer than one shoot. Observed SND is significant when the absolute value is greater than 1.96 (Epperson & Alvarez-Buylla 1997):

$$SND = (J_{\text{observed}} - \mu_1') / \sqrt{\mu_2}$$

where J_{observed} is the observed number of join genotypes; μ_1' is the expected number of joins for random distribution; and μ_2 is the variance.

In order to identify the gene flow ranges by allele, we calculated coancestry (r_{ij}) for the overall population and major mature trees separately (Reynolds *et al.* 1983; Heywood 1991; Loiselle *et al.* 1995):

$$r_{ij} = \frac{\sum_i (p_i - \bar{p})(p_j - \bar{p})}{k\bar{p}(1 - \bar{p})} + \frac{2}{\sqrt{8k + 1} - 1}$$

($i < j$)

$$k = \frac{n(n-1)}{2}$$

where p_i is the allele frequencies of each shoot (0, 0.5, or 1.0); and n is the number of shoots for the distance class.

The second term is a correction for finite sample size in which r_{ij} equals zero when the population fits the Hardy-Weinberg Equilibrium (HWE) (Loiselle *et al.* 1995). This parameter ranges from zero to one, and is parallel to

Wright's coefficient of relationships (Wright 1922). We followed the procedure of Loiselle *et al.* (1995) to test the significance.

Results

Genetic diversity for root sucker populations

The profiles of each study plot are summarized in Table 1. The number of shoots per square meter ranges from 0.38 in PWD to 1.40 in Q1. The number of polymorphic loci differs among study plots (Table 1). Powdermill shows polymorphisms in eight loci: *6Pgdh2*, *6Pgdh3*, *Acp2*, *Amy2*, *Fum*, *Lap*, *Mdh* and *Pgm*. Mont St. Hilaire shows polymorphisms in eight loci: *6Pgdh1*, *6Pgdh2*, *6Pgdh3*, *Acp1*, *Lap*, *Mdh*, *Pgi* and *Pgm*. Q1 shows polymorphisms in 11 loci: *6Pgdh2*, *6Pgdh3*, *Acp2*, *Adh1*, *Amy2*, *Fum*, *Got1*, *Lap*, *Mdh*, *Pgi* and *Pgm*. Q2 shows polymorphisms in nine loci: *6Pgdh2*, *6Pgdh3*, *Adh1*, *Amy2*, *Fum*, *Lap*, *Mdh*, *Pgi* and *Pgm*.

Size-class structures of shoots for each study plot are shown in Fig. 1a. It clearly shows that there is a peak in size-class 3 except for Q1. The number of multilocus genotypes for each size-class was approximately parallel to the number of shoots except for PWD, in which size-class 3 had fewer but size-class 5 had more multilocus genotypes compared to the number of shoots (Fig. 1b). The numbers of shoots per multilocus genotype are shown in Fig. 1c. Large numbers were observed in size-classes 3 and 4 for PWD, MSH, and Q2, indicating that shoots with identical genotypes are abundant in these intermediate size-classes. Uniqueness indices were high in size-class 5, except for MSH, in which size-class 2 indicated high levels of uniqueness (Fig. 1d). This index stands for the number of multilocus genotypes per shoot, suggesting that a rich variety of genotypes compared to the number of shoots was pooled in the mature size-class.

Differences in genetic diversity among size-classes

Differences in genetic variation among five size-classes were compared for genotype, allele and homozygote/heterozygote frequencies. The chi-squared test of independence showed significant values in the following criteria: genotypes and zygote frequencies of *Lap* and *6Pgdh2*, and zygotes of *Fum* (Fig. 2) in PWD; alleles of *Lap* and *Pgm* (Fig. 3) in MSH; and genotypes and zygotes of *Lap*, alleles of *6Pgdh2*, zygotes of *6Pgdh3* (Fig. 4), genotypes of *Pgm*, and genotypes, alleles, and zygotes of *Acp2* in Q1. There is no significance in the chi-squared test of Q2 (Fig. 5). There were rare genotypes and alleles in intermediate size-classes: *Lap* and *Pgm* in PWD; *6Pgdh1*, *6Pgdh3* and *Pgi* in MSH; *Got1* and *Pgm* in Q1; *Fum* in Q2. Also a specific variation was obtained in size-class 1 of *6Pgdh3* in PWD. The

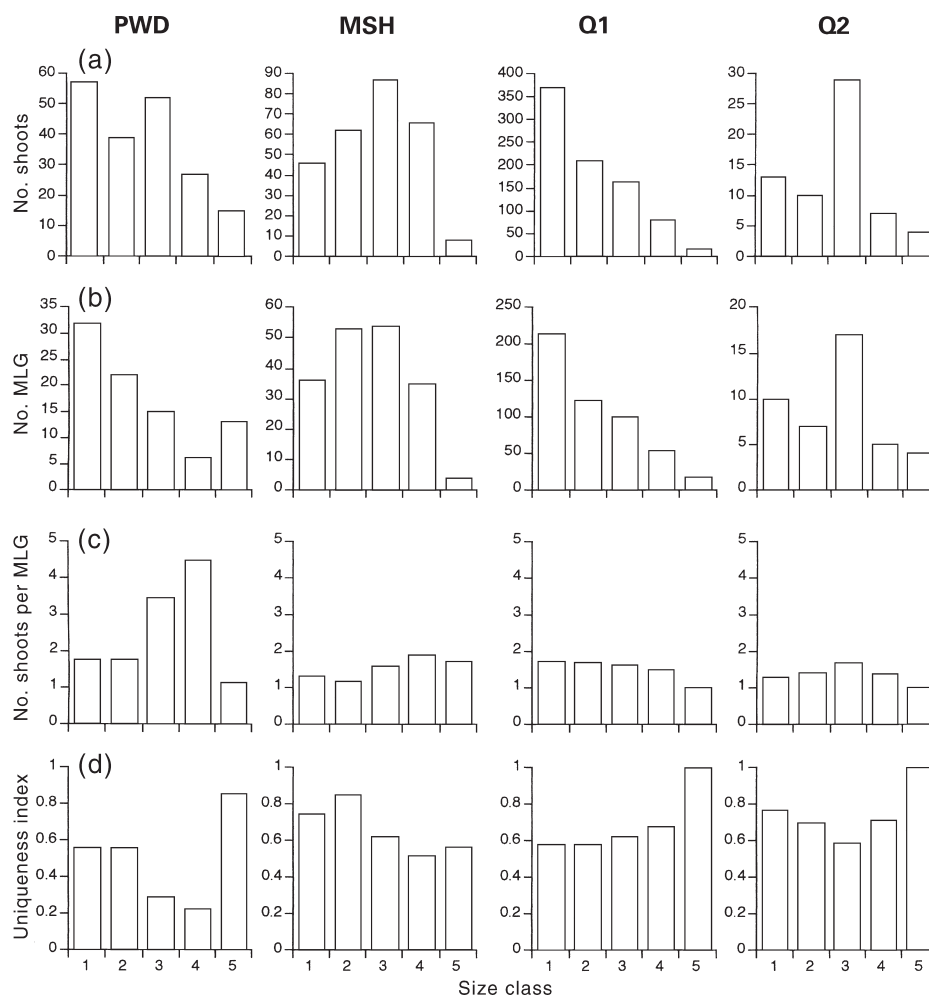


Fig. 1 Size-class distributions of four parameters related to the number of shoots and multilocus genotypes (MLG): (a) number of shoots; (b) number of MLG; (c) number of shoots per MLG; (d) number of MLG per shoot (uniqueness index).

existences of these rare genetic variations in certain size-classes were not detected at all by the chi-squared test.

The above-mentioned polymorphic loci were used to identify multilocus genotypes for each study plot. The average numbers of shoots per multilocus genotype varied from 1.17 in Q2 to 3.22 in PWD. Simpson's indices (D) were high for all study plots, which range from 0.91 in PWD to 0.99 in Q1 and Q2. Fager's indices of evenness (E) also showed a high level, ranging from 0.83 in PWD to 0.98 in Q1.

Table 2 shows clonal diversity measures overall and for each size-class. Both Simpson's index (D) and Fager's genotypic evenness (E) for overall size-classes showed high diversity for all study plots. Because of the small number of samples, size-class 5 did not give reliable E measures for all study plots. In PWD and Q1, both D and E were low in size-classes 3 and 4. In MSH, D was low for size-class 5, and E was low for size-classes 2 and 3. In Q2, D was low in size-class 4 and E was low in size-class 2. The result of E at size-class 4 in Q2 was zero, because the observed D -value equals the minimum value.

F-statistics

Wright's F -statistics were calculated to analyze genetic differentiation among size-classes. Table 3 summarizes results of calculations, in which F_{IT} and F_{IS} show positive significant values in all study plots. F_{ST} shows significant values at PWD and Q1, indicating the genetic differentiation among size-classes. F_{IS} values for each size-class are shown in Fig. 6. Positive significant values for F_{IS} were observed in the intermediate size-classes: size-classes 3 and 4 in PWD, classes 2 and 3 in MSH, and classes 3 and 4 in Q1.

Spatial distributions of multilocus genotypes

The numbers of multilocus genotypes detected were 59, 136, 354, and 54 at PWD, MSH, Q1, and Q2, respectively (Table 1). Distributions of these multilocus genotypes were mapped as shown in Fig. 7. These maps clearly show extension and intermingling of each genotype (namely, ramets of the identical genet) except for Q2, where the

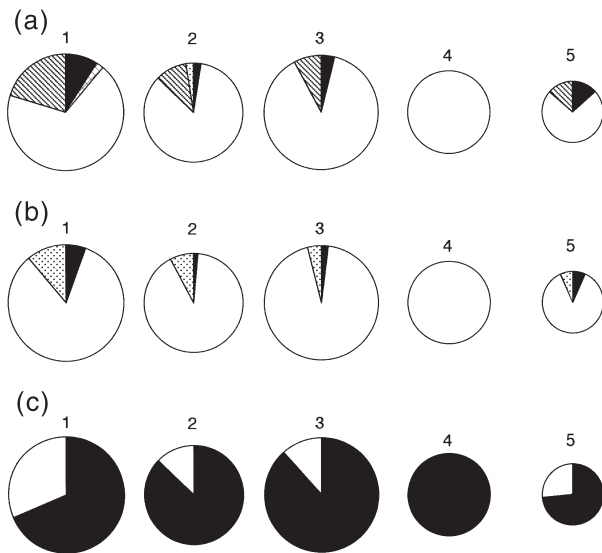


Fig. 2 Genotype, allele and homozygote and heterozygote frequencies of *Fum* at PWD: (a) ■ *ab*, ▨ *ac*, □ *bb*, ▩ *bc*, ▤ *cc*; (b) ■ *a*, □ *b*, ▨ *c*; (c) ■ homozygote, □ heterozygote. Numbers 1–5 refer to five size-classes described in the text. The size of each diagram represents the relative number of shoots included.

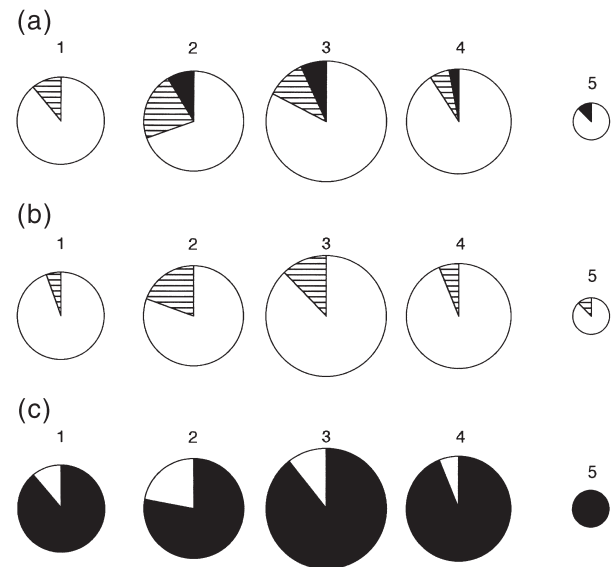


Fig. 3 Genotype, allele and homozygote and heterozygote frequencies of *Pgm* at MSH: (a) □ *aa*, ▨ *ab*, ■ *bb*; (b) □ *a*, ▨ *b*; (c) ■ homozygote, □ heterozygote. Numbers 1–5 refer to five size-classes described in the text. The size of each diagram represents the relative number of shoots included.

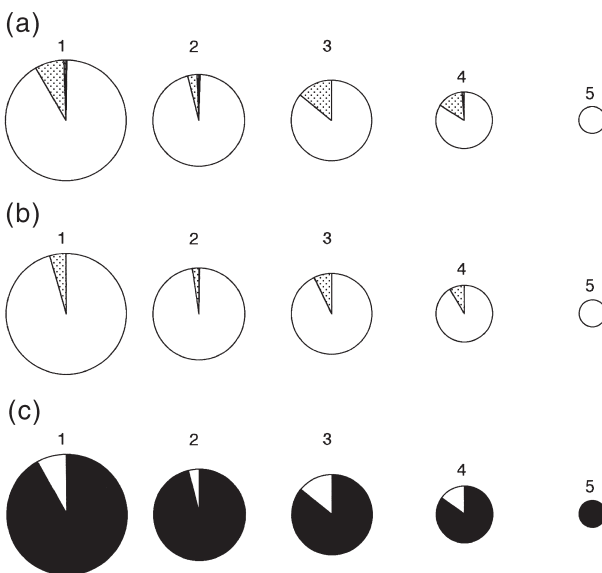


Fig. 4 Genotype, allele and homozygote and heterozygote frequencies of *6Pgdh3* at Q1: (a) ■ *ab*, □ *bb*, ▨ *bc*, ▩ *cc*; (b) ■ *a*, □ *b*, ▨ *c*; (c) ■ homozygote, □ heterozygote. Numbers 1–5 refer to five size-classes described in the text. The size of each diagram represents the relative number of shoots included.

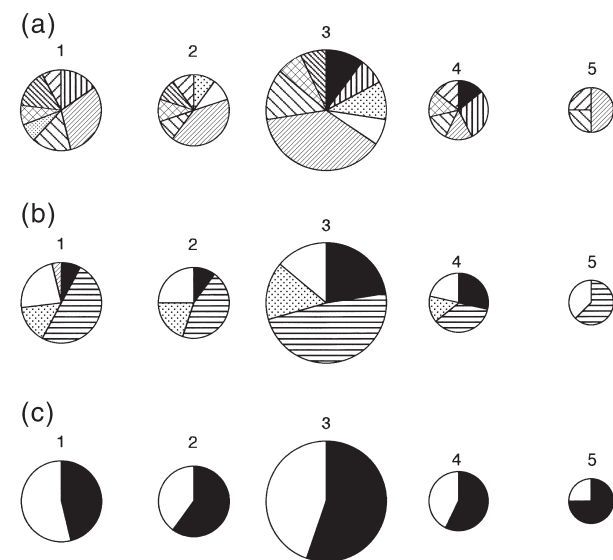


Fig. 5 Genotype, allele and homozygote and heterozygote frequencies of *Lap* at Q2: (a) ■ *aa*, ▨ *ad*, □ *cc*, ▩ *cf*, ▤ *de*, ▨ *ac*, □ *ae*, ▨ *ce*, ▩ *dd*, ▨ *ee*; (b) ■ *a*, □ *c*, ▨ *d*, □ *e*; (c) ■ homozygote, □ heterozygote. Numbers 1–5 refer to five size-classes described in the text. The size of each diagram represents the relative number of shoots included.

American beech is extremely suppressed by the sugar maple, *A. saccharum*.

In PWD (Fig. 7a), four large genets were found to occur within the study plot with various genotypes of small-sized shoots. Surrounding mature shoots have different

multilocus genotypes from those in the interior of the study plot. The four major genets are distributed in clusters and separated from each other.

Particularly in MSH for multilocus genotypes with large-sized shoots, the distribution map clearly shows

Population	PWD		MSH		Q1		Q2	
Size-class	<i>D</i>	<i>E</i>	<i>D</i>	<i>E</i>	<i>D</i>	<i>E</i>	<i>D</i>	<i>E</i>
1	0.95	0.82	0.99	0.89	0.99	0.97	0.95	0.49
2	0.95	0.83	0.99	0.75	0.99	0.94	0.91	0.52
3	0.77	0.63	0.96	0.75	0.98	0.91	0.95	0.84
4	0.66	0.60	0.96	0.87	0.98	0.91	0.86	0.00
5	0.98	0.54	0.81	0.59	1.00	1.00	1.00	1.00
Overall	0.91	0.83	0.97	0.89	0.99	0.98	0.99	0.85

Table 2 Two measures of clonal diversity, Simpson's index (*D*) and Fager's genotypic evenness (*E*), for each size-class

Study plot	<i>F_{IT}</i>	<i>F_{ST}</i>	<i>F_{IS}</i>	<i>R</i>
PWD	0.086 (0.150)*	0.019 (0.006)**	0.069 (0.158)*	0.033 (0.016)
MSH	0.063 (0.073)**	0.005 (0.004)	0.058 (0.073)**	0.009 (0.008)
Q1	0.049 (0.037)**	0.004 (0.001)**	0.045 (0.037)**	0.008 (0.002)
Q2	0.105 (0.115)*	-0.014 (0.008)	0.118 (0.119)**	-0.027 (0.014)

Table 3 Wright's *F*-statistics and Hamilton's relatedness for five size-classes at four study plots

Standard errors obtained from jackknifing method are given in parentheses.

Test of significance by permutation for *F*-statistics: **P* < 0.05, ***P* < 0.01.

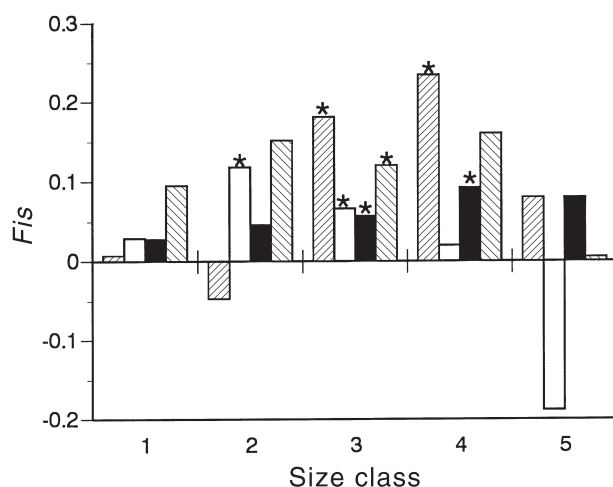


Fig. 6 Inbreeding coefficient (*F_{IS}*) for each size-class: ▨ PWD; □, MSH; ■, Q1; ▤ Q2. *Significantly different from zero, *P* < 0.05.

radial patterns in the root sucker formations from large shoots (Fig. 7b). Root suckers of large-sized shoots are intermingling with each other, while genets of the small-sized shoots are clustered. There was a new gap in the left half (small x-coordinate) of the study plot where unique genotypes were abundant. In contrast, the right side (large x-coordinate) of the study plot, where the old gap existed, was occupied by a limited number of multilocus genotypes.

Since Q1 (Fig. 7d,e) is a large plot on a steep slope, the extensions of the root sucker are rather small at the top (smaller x-coordinate) compared to PWD and MSH, while the radial patterns of root sucker extensions are obvious at the bottom, where the slope is gentle. It is notable that not only root suckering genotypes but also miscellaneous multilocus genotypes with small to intermediate size-classes are intermingled. The surrounding large trees have specific multilocus genotypes, which are different from those in the interior of the study plot.

Q2 is established at the base of the slope of Q1, where the American beech is extremely suppressed by the sugar maple and shoot density of the American beech is lowest of all four study plots (Fig. 7c). No root sucker extension is evident. Various multilocus genotypes are scattered within a dense thicket of sugar maple juveniles. Several genotypes share more than one shoot, but they are not connected to large-sized shoots, which have the potential to produce root suckers.

Spatial autocorrelation analyses

Standard normal deviate. Results of the *SND* calculations for like-joins at PWD are demonstrated in Fig. 8. It is obvious that smaller distance classes show positive relationships and larger classes show negative values. All of the positive significant values (larger than 1.96) are observed in distance classes of 5, 10 and 15 m, and all of the negative significant values (smaller than -1.96) in distance classes of 15, 20, 25, and 30 m. The *SND* values

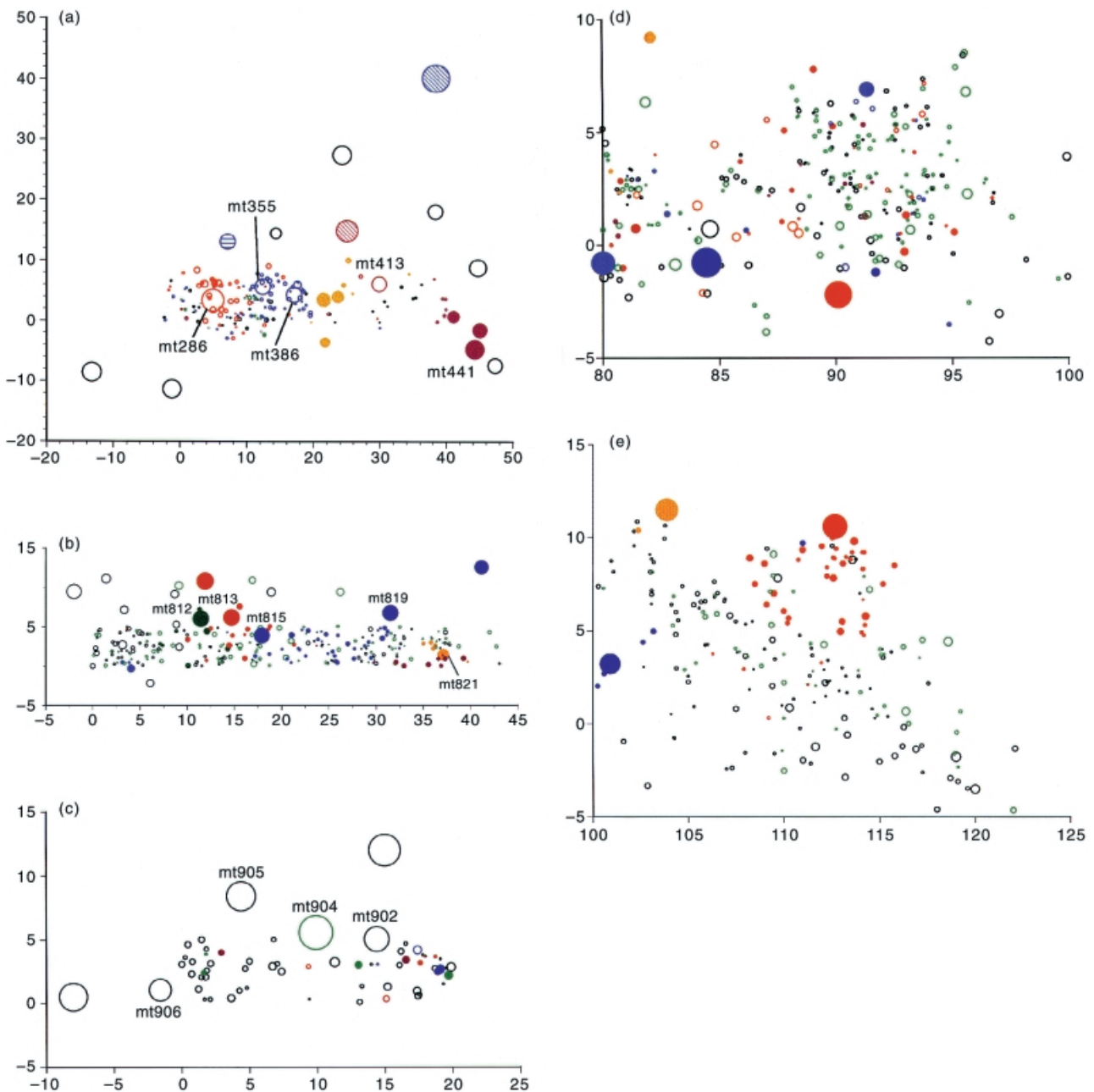


Fig. 7 Map of multilocus genotypes with mother tree identification number for coancestry calculations: (a) PWD; (b) MSH; (c) Q2; (d) Q1 (80 m < x < 100 m); (e) Q1 (100 m < x). Scales are in meters. Size of a circle is relative to the diameter of a shoot. Black circles represent shoots with unique multilocus genotypes. Different symbols refer to different multilocus genotypes.

change from positive to negative at distance classes of 15–20 m. From these results, identical genotypes seem to aggregate within a 15–20 m radius in the PWD population.

Almost the same tendencies were observed in MSH (Fig. 9), where positive relationships cluster in smaller distance classes. Compared to PWD, the negative relationships in larger distance classes were not significant

in MSH. Significant values were observed only in *Lap-cc* and *-ce*. Positive aggregation of genotypes is suggested within a 15–20 m radius.

In Q1, we divided the study plot into two sections on the upper and lower parts of the slope. At the upper part of the slope, which is steeper and partly covered by *Taxus canadensis*, no significant *SND* values were obtained (Fig. 10). Several genotypes show positive values in

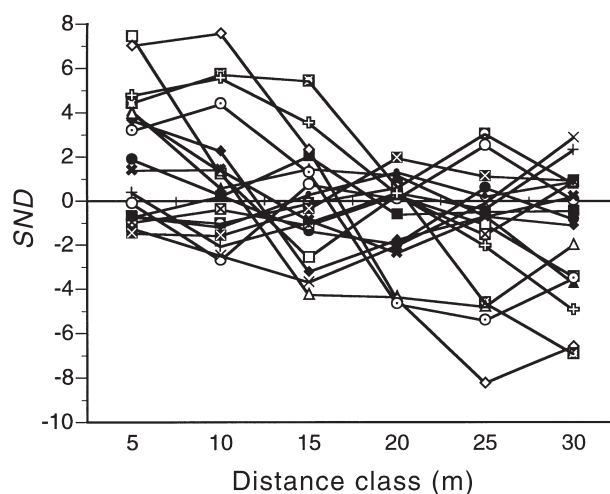


Fig. 8 Standard normal deviate (SND) for the transect at Powdermill, PA, USA. $\square$, 6Pgdh2-aa; $\circ$, 6Pgdh2-ab; $\triangle$, 6Pgdh2-bb; $\diamond$, 6Pgdh3-bb; $\boxtimes$, 6Pgdh3-bc; $\times$, Acp2-aa; $\bullet$, Amy2-aa; $\boxminus$, Fum-ab; $\ominus$, Fum-bb; $\square$, Fum-bc; $\blacksquare$, Lap-ae; $\bullet$, Lap-cc; $\blacktriangle$, Lap-ce; $\blacklozenge$, Lap-ee; $+$, Mdh-ab; $\oplus$, Mdh-bb; $\square$, Pgm-bb; $\times$ Pgm-bc.

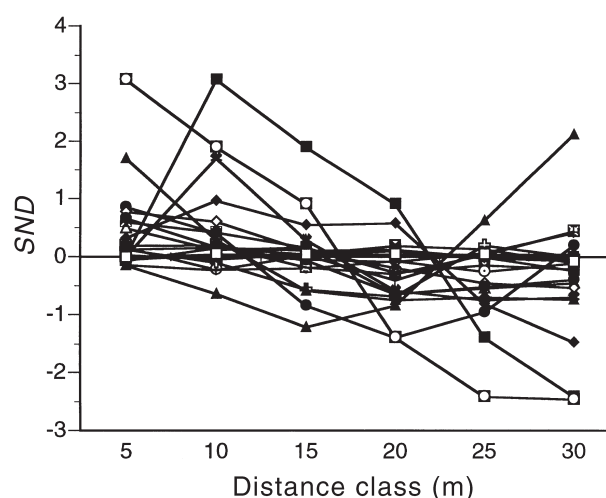


Fig. 9 Standard normal deviate (SND) for the transect at Mont St. Hilaire, Quebec, Canada. $\triangle$, 6Pgdh1-aa; $\diamond$, 6Pgdh1-ab; $\boxtimes$, 6Pgdh2-aa; $\boxminus$, 6Pgdh2-ab; $\ominus$, 6Pgdh2-bb; $\square$, 6Pgdh3-bb; $\square$, 6Pgdh3-bc; $\times$, Acp1-aa; $\star$, Acp1-ab; $\bullet$, Acp1-bc; $\blacktriangle$, Acp1-cc; $\blacksquare$, Lap-cc; $\bullet$, Lap-ce; $\blacktriangle$, Lap-cf; $\blacklozenge$, Lap-de; $\square$, Lap-ef; $\circ$, Lap-ff; $\blacklozenge$, Mdh-ab; $\square$, Mdh-bb; $\times$, Pgi-bb; $\bullet$, Pgi-bd; $\oplus$, Pgi-dd; $\times$, Pgm-aa; $\oplus$, Pgm-bb; $\oplus$, Pgm-ab; $+$, Pgm-bb.

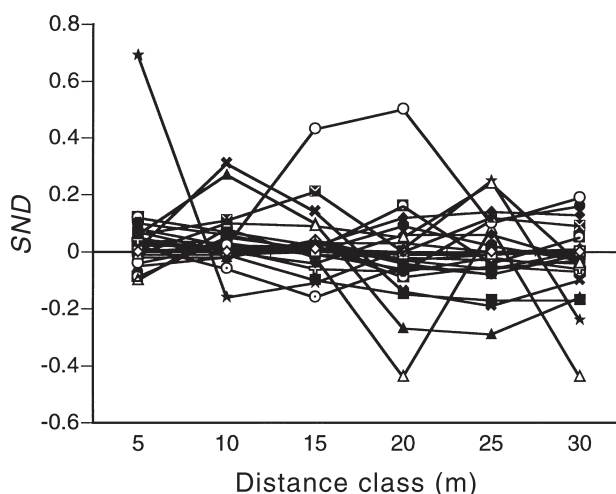


Fig. 10 Standard normal deviate (SND) for the upper slope transect in Duchesnay, Quebec (Q1). $\triangle$, 6Pgdh2-aa; $\diamond$, 6Pgdh2-ab; $\boxtimes$, 6Pgdh2-bb; $\boxminus$, 6Pgdh3-bb; $\ominus$, 6Pgdh3-bc; $\square$, Acp2-aa; $\circ$, Acp2-ab; $\triangle$, Acp2-bb; $\star$, Adh-ab; $\blacksquare$, Adh-bb; $\diamond$, Amy2-aa; $\boxtimes$, Amy2-ab; $\square$, Fum-ab; $\times$, Fum-bb; $\times$, Fum-bc; $\blacksquare$, Lap-aa; $\bullet$, Lap-ac; $\blacktriangle$, Lap-ae; $\blacklozenge$, Lap-cc; $\square$, Lap-ce; $\circ$, Lap-ee; $\bullet$, Mdh-aa; $\blacktriangle$, Mdh-ab; $\blacklozenge$, Mdh-bb; $\bullet$, Pgi-bb; $\oplus$, Pgi-be; $\times$, Pgi-ee; $+$, Pgm-bb; $\oplus$, Pgm-bc; $\times$, Pgm-cc.

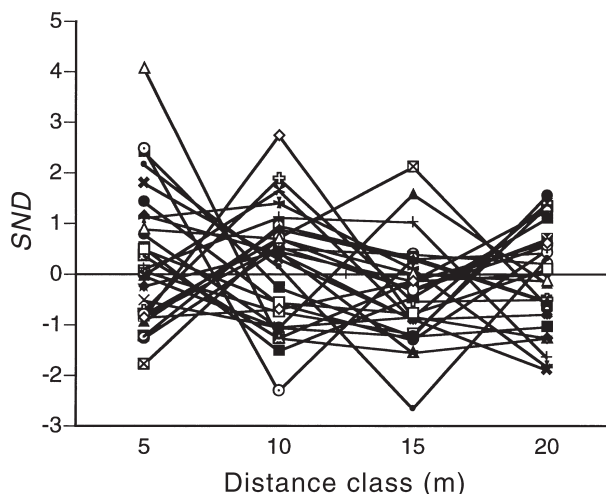


Fig. 11 Standard normal deviate (SND) for the second transect in Duchesnay, Quebec (Q2). $\boxtimes$, 6Pgdh2-ab; $\ominus$, 6Pgdh2-bb; $\square$, 6Pgdh3-bb; $\blacktriangle$, Adh1-bb; $\triangle$, Amy2-aa; $\diamond$, Amy2-ab; $\square$, Fum-ab; $\times$, Fum-bb; $+$, Fum-bc; $\oplus$, Fum-bc; $\blacksquare$, Lap-aa; $\bullet$, Lap-ac; $\blacktriangle$, Lap-ad; $\blacklozenge$, Lap-ae; $\square$, Lap-cc; $\circ$, Lap-ce; $\triangle$, Lap-dd; $\diamond$, Lap-de; $\boxtimes$, Lap-ee; $\blacklozenge$, Mdh-aa; $\square$, Mdh-ab; $\circ$, Mdh-bb; $\oplus$, Pgi-bb; $\times$, Pgi-bc; $\star$, Pgi-bd; $\blacksquare$, Pgi-be; $\bullet$, Pgi-cc; $\times$, Pgm-bb; $\bullet$, Pgm-bc.

smaller distance classes and negative values in larger classes, although the patterns are contrary to another couple of genotypes. No significant value was observed at the lower part of the slope, either. Trends toward aggregation of identical genotypes were observed in 6Pgdh2.

In Q2, calculations for SND were carried out up to the distance class of 20 m, because of the small size of

the study plot (Fig. 11). Positive significant values were observed in the smaller distance classes 5, 10 and 15 m, and negative significances were observed in 10 and 15 m.

Coancestry (r_{ij}). The results of coancestry calculations for the overall study plot are shown in Fig. 12. It is obvious for all study plots that the smaller distance classes

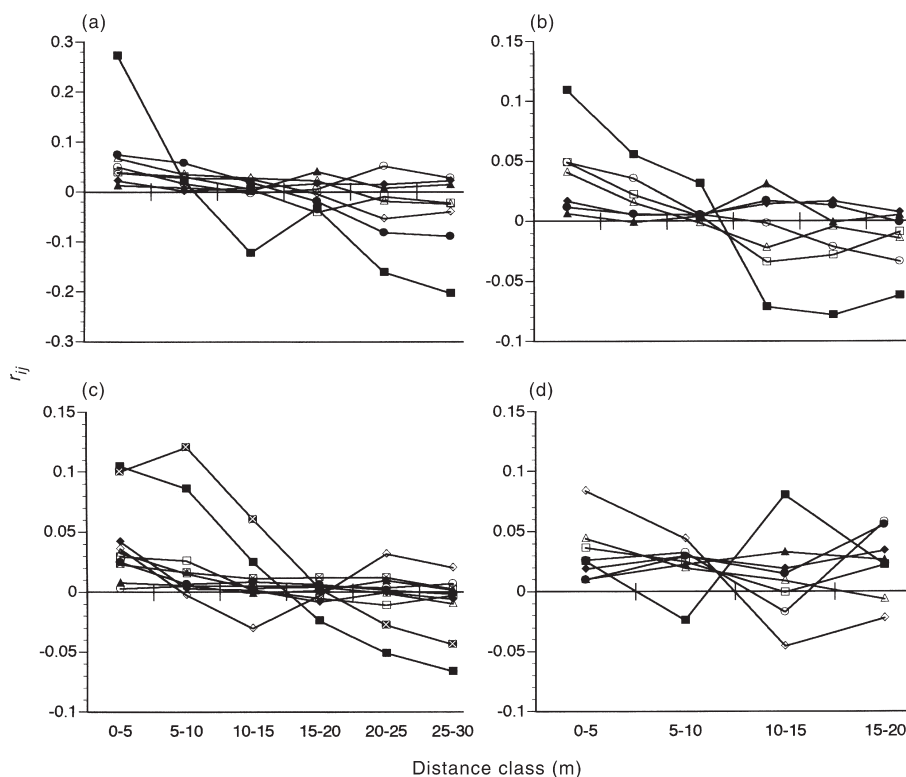


Fig. 12 Coancestry (r_{ij}) estimation for overall study plot. (a) PWD, (b) MSH, (c) Q1, (d) Q2. See Appendix I for significant values. (a) ■, 6Pgdh2; ●, 6Pgdh3; ▲, Acp2; ◆, Amy3; □, Fum; ○, Lap; △, Mdh; ◇, Pgm; (b) ■, 6Pgdh1; ●, 6Pgdh2; ▲, 6Pgdh3; ◆, Fum; □, Lap; ○, Pgi; △, Pgm; (c) ■, 6Pgdh2; ●, 6Pgdh3; ▲, Acp2; ◆, Adh1; □, Amy2; ○, Fum; △, Lap; ◇, Mdh; ✕, Pgi; ✕, Pgm; (d) ■, 6Pgdh2; ●, 6Pgdh3; ▲, Amy2; ◆, Fum; □, Lap; ○, Mdh; △, Pgi; ◇, Pgm.

showed positive, but larger classes showed negative, relationships. The tests of significance also supported these results (Appendix I). In PWD, positive significant values were observed in distance classes 0–5 m and 5–10 m, while negative significance was observed in classes 10–15 m, 15–20 m, 20–25 m and 25–30 m. In MSH, positive significance was found in classes 0–5 m, 5–10 m and 10–15 m, while negative significance was observed in classes 15–20 m, 20–25 m and 25–30 m. In Q1, positive significance was found in classes 0–5 m, 5–10 m and 10–15 m, and negative significance in classes 10–15 m, 15–20 m, 20–25 m and 25–30 m. In addition, *Mdh* showed positive significance in classes 20–25 m and 25–30 m in Q1. In Q2, only *Pgm* showed significant positive values in class 0–5 m, but it was negative in 10–15 m.

Coancestry calculations for major mature trees appeared on multilocus genotype maps (Fig. 7) are summarized in Appendices II–IV.

In PWD, positive relationships with major mature trees were evident from the multilocus genotype map (Fig. 7a). Five mother trees had positive significant values in distance class 0–5 m; two of them showed positive significance at classes 5–10 m and 10–15 m, while most of the negative significance was observed in class 20–25 m and 25–30 m (Appendix II).

Similar trends of coancestries were observed in four mature trees in MSH, which showed positive relationships within a 10 m distance and negative for a greater dis-

tance of 15 m. Significant values, however, were observed in one locus each for four mature trees (Appendix III). The exception for mt815 indicates no correlated distribution of identical genotypes at a distance of 0–25 m, and seems to skew positive in distance class 25–30 m. However, the r_{ij} for mt815 was not statistically significant at all.

Also in Q1, similar patterns for correlograms (positive in smaller distance classes and negative in larger classes) were observed for major mature trees (Appendix IV). Positive significant values were observed up to 15 m distance classes, and most of the negative significance was at the farther distance classes 15–30 m.

For the Q2 study plot, coancestries were calculated up to 20 m distance. Correlograms for mt902 and mt904 seem to have positive relationships in distance class 15–20 m, and mt905 shows a positive relationship in 0–10 m distance classes. However, no significant value was observed in Q2.

Discussion

Implication of root sucker formation for temporal maintenance of genotypic variability within local populations

Increasing numbers of studies during the last decades have revealed that clonal plants have the same levels of genetic diversity as non-clonal plants (Ellstrand & Roose

1987; McLellan *et al.* 1997; Kudoh *et al.* 1999; S. Kawano *et al.*, unpublished observation on *Uvularia perfoliata*, 1997). The American beech, too, has the same level of genetic diversity regardless of whether the population shows vegetative reproduction by root sucker formation or not (K. Kitamura & S. Kawano, unpublished observation, 1998). If the vegetatively reproduced shoots dominate in the population, genetic variability seems to become low; however, genetic analyses of clonal plant populations in the past decade have revealed that the polymorphisms are maintained because genotypes are pooled within a population for more than its shoot lifetime (Ellstrand & Roose 1987; Hamrick & Godt 1989; Widen *et al.* 1994). Species with asexual clonal reproduction often possess considerable variation within and between local populations (Ellstrand & Roose 1987; S. Kawano *et al.*, unpublished observation, 1997).

The present study revealed that the clonal diversity of the American beech with root sucker formation ranges from 0.91 to 0.99 for Simpson's index (D), and from 0.83 to 0.98 for Fager's index of evenness (E) (Table 1). These estimations are considerably high relative to the reports for other clonal species (see reviews of Ellstrand & Roose 1987; Widen *et al.* 1994). This level of clonal diversity is similar to other woody clonal species such as *Populus tremuloides* (Cheliak & Dancik 1982; Jelinski & Cheliak 1992), *Betula glandulosa* (Hermanutz *et al.* 1989), and *Quercus chrysolepis* (Montalvo *et al.* 1997).

Most of the chi-squared tests of independence for genotypes, alleles, and zygotes showed no significance among five size-classes or specific trend in each study plot. This result of statistical testing, however, should be interpreted carefully, because slight differences in sampling scales, numbers of shoots, and characteristics of each locus are not always detectable.

Values of F_{IT} and F_{IS} were significantly positive in all populations, suggesting deviations from HWE because of the vegetative reproduction by root sucker formation. The implications of root sucker formation for temporal genetic substructuring may be regarded in terms of genetic reservoirs for a local population, which keeps genotypes once adapted to the environment of the established site. Root sucker formation obviously contributes to the number of shoots in intermediate size-classes (Fig. 1). These genotypes were derived from various generations of sexual reproduction. Evidence for the overlapping generations are demonstrated in F_{IS} for each size-class. Deviations from HWE were observed as positive significant values for F_{IS} in intermediate size-classes. These constitute the evidence of overlapping generations for intermediate size-classes.

Also these intermediate size-classes, which were positively significant in F_{IS} , were identical to the low estimates in E measures except for Q2, size-classes 3 and 4 in PWD,

classes 2 and 3 in MSH, and classes 3 and 4 in Q1. The low estimates in E also suggested the unevenness of genotype frequencies, which can be considered to mean that less frequent genotypes are pooled in these size-classes. It is notable that there were several rare genotypes, which are unique to a certain size-class (Figs. 2–5, 7). The newly established seedlings can introduce unique genotypes into local populations, as is evident in the genetic structures of size-class 1. The rare genotypes in intermediate size-classes can be interpreted as remnant shoots derived from dead individual shoots that previously lived in the sampled area. This situation is significant, especially in Q2 (Fig. 7c).

Ranges of root sucker extension and aggregation intensity of genotypes

For the individual genet, root suckering is a means to occupy space and increase in size. We attempted in the present study indirect estimations of genet size by multilocus genotypes, which are regarded as suitable genetic markers to detect fine-scale spatial structures of clonal plant populations (Widen *et al.* 1994).

The maps of the multilocus genotypes (Fig. 7) demonstrated spatial distributions of genets. The detected multilocus genotypes suggested that all of the study plots were multiclonal populations (Fig. 7). The multilocus genotype maps revealed that there are well-developed genetic structures by root sucker formation in PWD, MSH, and Q1, but only in the suppressed site of Q2. The maps of PWD, MSH, and Q1 (Fig. 7a,b,e) clearly show radial patterns of extension of root suckers. Correlograms of SND and r_{ij} supported this evidence concerning the manner of isolation by distance (Hamrick & Nason 1996). Genet intermingling is obvious, and the branching angle of roots is not uni-directional. Jones and Raynal (1987) revealed that the shoots, which begin producing root suckers, are approximately 20 cm in diameter. The size advantage in production of root suckers (Jones & Raynal 1987) was also evident in the fact that large shoots are attributed to dominant genotypes. Spatial autocorrelations (r_{ij}) for each mature shoot also demonstrated a remarkable fine-scale spatial genotype structure.

Epperson (1995) mentioned that the SND is more informative when it is applied to joins of homozygotes. In order to determine the extension range of root sucker shoots, we consider joins of heterozygotes to be informative as well. For the same reason, we simply calculated like-joins of genotypes. For all study plots, spatial autocorrelation estimates turn from positive to negative values at 15–20 m distance classes. These results suggest that the 15–20 m distance is a ramet radial by root sucker formation for the American beech. Half of this distance (7.5–10 m) can be suggested as the root sucker extensions,

which is similar to the 8 m estimate in the field survey (Jones & Raynal 1986).

Levels of intensity of root suckering differ among loci and populations. Co-ancestry (r_{ij}), which is parallel to the inbreeding coefficient, shows clear trends for aggregation of ramets within 15 m distance (Fig. 12; Appendix I). Although the theoretical inbreeding coefficient of half-sibs is 0.25, most of the empirical values in the 0–5 m distance class range within 0–0.1. This may be due to overlapping root suckers of the intermingling diverse genets on a fine spatial scale.

Significant relatedness in co-ancestry analyses for both overall population and each mature shoot was observed within a distance class of 15 m. These results were identical to *SND*, suggesting that the genetic substructurings were attributed mainly to the vegetative reproduction by root sucker formation rather than sexual reproduction.

Implications of root suckers for the spatio-temporal genetic structure of northern populations

The American beech begins seed production at approximately 40 years of age (Fowells 1965). The age for root sucker production is earlier than seed production age (Jones & Raynal 1987). Root suckering in the American beech populations is intensive under harsher environmental conditions (Held 1983). Our previous study on the geographic variations of the American beech (K. Kitamura & S. Kawano, unpublished observation, 1998) suggested genetic similarity among root sucker populations, which had retreated to the territories glaciated during the post-Pleistocene period. In these glaciated environments, characterized by shallow soils and exposed roots, active root sucker formation by specific genotypes, possibly adapted to that site, is assumed to be advantageous for maintenance of local populations.

Root sucker formation has a significant role in preserving genetic diversity in intermediate size-classes, in addition to the rare genetic variations produced and supplied to small size-classes via pollen flows (Alvarez-Buylla & Garay 1994; Kitamura *et al.* 1997b). Vegetatively reproduced shoots can maintain the genet over more than the life-span of a single shoot, to maintain particular adapted genotypes within the population for a longer period of time (Kitamura *et al.* 1992, 1997a,b; Kawano & Kitamura 1997).

Simultaneously, root sucker formation renders the genet capable of exploiting gaps or surrounding spaces of the mature trees and of capturing more resources in the habitats, where intensive competitive exclusion may be in operation among some other woodland elements; thus this growth habit is advantageous for occupying space around its establishment site. The fine-scale genetic sub-

structurings within local populations could be affected more by root sucker formation (McCaig 1993) than by establishment of new sexually recruited individuals (Berg & Hamrick 1995). Dominant genotypes may keep their advantages spatially as well as temporally in such changing environments. In the northern populations with root sucker formation, seedling establishment also plays a significant role in the recruitment of the succeeding generations, and as a consequence forming multiclonal local populations.

Some more critical demographic genetic analyses of the remnant root sucker populations in the Blue Ridges and Appalachian Mountains are now in progress, and will shed light upon the differentiation patterns of local populations of the American beech (K. Kitamura *et al.*, unpublished data, 1998).

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Appendix I.

Significant values of r_{ij} for each distance class (see Fig. 12).

Study plot	Locus	Distance class					
		0–5 m	5–10 m	10–15 m	15–20 m	20–25 m	25–30 m
PWD	6Pgdh2	+	•	–	•	–	–
	6Pgdh3	+	+	•	–	–	–
	Fum	•	•	•	•	•	•
	Mdh	+	+	•	•	–	–
	Pgm	•	•	•	–	•	•
MSH	6Pgdh1	+	+	+	–	–	–
	Lap	+	•	•	–	–	•
	Pgi	+	+	•	•	–	–
	Pgm	•	•	•	–	•	•
Q1	6Pgdh2	+	+	+	–	–	–
	6Pgdh3	+	•	•	•	•	•
	Adh1	+	•	•	–	•	–
	Amy2	+	+	•	–	–	–
	Lap	+	+	•	•	•	–
	Mdh	+	•	–	•	+	+
	Pgi	+	+	+	•	–	–
Q2	Pgm	+	+	•	•	•	•
	Pgm	+	•	–	•	•	•

+, Positive significant value at a given distance class; –, negative significant value; •, not significant.

Appendix II.

Significant values of r_{ij} for each mother tree at PWD.

Mother tree	Locus	Distance class					
		0–5 m	5–10 m	10–15 m	15–20 m	20–25 m	25–30 m
mt286	6Pgdh2	+	•	–	•	•	•
	6Pgdh3	+	•	•	•	•	•
	Acp2	+	•	•	•	•	•
	Lap	+	•	•	•	•	•
	Pgm	•	•	+	•	•	•
mt355	6Pgdh2	+	•	•	•	•	•
	6Pgdh3	+	•	•	•	•	•
	Mdh	•	•	•	•	•	–
	Pgm	•	•	•	•	–	–
mt386	6Pgdh2	+	•	–	•	•	•
	Lap	•	•	•	•	+	•
	Pgm	+	•	•	•	–	•
mt413	6Pgdh2	•	•	+	•	–	–
	6Pgdh3	•	+	•	•	•	•
	Fum	+	+	•	•	–	•
	Lap	•	•	+	•	•	•
mt441	6Pgdh2	•	•	•	•	•	–
	6Pgdh3	•	+	•	•	+	•
	Fum	+	+	•	•	•	•
	Lap	•	+	•	•	•	+
	Mdh	•	+	•	•	•	•
	Pgm	•	+	•	•	•	–

+, Positive significant value at a given distance class; –, negative significant value; •, not significant.

Appendix III.

Significant values of r_{ij} for each mother tree at MSH.

Mother tree	Locus	Distance class					
		0–5 m	5–10 m	10–15 m	15–20 m	20–25 m	25–30 m
mt812	6Pgdh1	•	•	•	•	–	•
mt813	6Pgdh1	•	+	•	–	•	•
mt819	6Pgdh1	+	+	•	–	•	•
mt821	Pgm	+	•	•	•	•	•

+, Positive significant value at a given distance class; –, negative significant value; •, not significant.

Appendix IV.*Significant values of r_{ij} for each mother tree at Q1.*

Mother tree	Locus	Distance class					
		0–5 m	5–10 m	10–15 m	15–20 m	20–25 m	25–30 m
mt1	<i>6Pgdlh1</i>	+	–	•	•	•	•
	<i>Lap</i>	+	+	•	•	•	•
	<i>Pgi</i>	+	•	•	•	•	•
	<i>Pgm</i>	+	•	•	•	•	•
mt116	<i>6Pgdlh2</i>	•	•	•	•	•	–
	<i>Adh1</i>	•	•	–	•	•	•
	<i>Lap</i>	+	+	•	•	•	•
	<i>Mdh</i>	•	•	•	•	•	–
	<i>Pgi</i>	•	•	+	•	•	•
	<i>Pgm</i>	•	+	+	•	•	•
mt331	<i>6Pgdlh2</i>	+	+	•	•	•	–
	<i>Adh1</i>	•	•	•	–	–	•
	<i>Amy2</i>	•	•	•	•	–	–
	<i>Mdh</i>	+	•	•	•	•	•
	<i>Pgi</i>	+	•	•	•	•	–
	<i>Pgm</i>	+	•	•	•	•	•
mt465	<i>6Pgdlh2</i>	+	+	+	•	–	–
	<i>6Pgdlh3</i>	•	+	•	•	•	•
	<i>Mdh</i>	•	•	+	•	•	•
	<i>Pgi</i>	•	•	•	–	–	–
	<i>Pgm</i>	+	•	•	•	•	•
	<i>Pgm</i>	+	•	•	•	•	•
mt607	<i>6Pgdlh2</i>	•	•	•	•	–	•
	<i>6Pgdlh3</i>	•	–	•	•	•	•
	<i>Mdh</i>	+	•	–	•	•	•
	<i>Pgi</i>	•	–	–	•	•	•
	<i>Pgm</i>	•	•	–	•	•	•
	<i>Pgm</i>	•	•	–	•	•	•
mt650	<i>6Pgdlh2</i>	•	+	+	•	–	–
	<i>Amy2</i>	+	+	•	•	•	•
	<i>Lap</i>	+	•	•	•	•	•
	<i>Mdh</i>	+	•	•	–	•	+
	<i>Pgi</i>	•	+	+	•	•	–
	<i>Pgm</i>	•	+	+	•	•	•
mt793	<i>6Pgdlh2</i>	+	+	•	•	•	•
	<i>Amy2</i>	•	–	–	•	•	•
	<i>Lap</i>	+	•	•	•	•	•
	<i>Mdh</i>	+	•	•	•	•	+
	<i>Pgi</i>	+	+	+	•	•	•
	<i>Pgm</i>	+	+	•	•	+	•
mt833	<i>6Pgdlh2</i>	•	–	–	–	•	•
	<i>6Pgdlh3</i>	–	•	•	•	+	•
	<i>Acp2</i>	•	•	•	•	•	+
	<i>Amy2</i>	•	•	•	–	•	•
	<i>Lap</i>	•	•	–	•	•	•
	<i>Pgi</i>	•	+	+	+	•	•
	<i>Pgm</i>	•	•	•	+	+	•
	<i>Pgm</i>	•	•	•	+	+	•

+, Positive significant value at a given distance class; –, negative significant value; •, not significant.

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